```
import json
import numpy as np



IMAGE_SIZE = (224, 224)

# Configuration


# Disease Info
```

```
from
impor
CONFI
BASE_
DISEA
```

```
# Multi-language translations
                                                                TRANS
```

```
def save_voice_file(text, language, output_path=OUTPUT_AUDIO_PATH):                                                
```

```
@st.cache_data
def l

def preprocess_image(image: Image.Image) -> np.ndarray:
    "

def predict(image_array: np.ndarray, model, class_names):
    "
```

```
    # Get best prediction index
                                                                                b
```

```
def main():
                                                                                #
```

```
/* Unique Icon Block Cards */

.confidence-card {

.severity-card {

.description-card {
```

```
.treatment-card {

.disease-name {

/* Enhanced Confidence Meter */

/* Color-coded severity levels */
```

```
@keyframes pulse {


/* Enhanced info cards with better visibility */


/* Status cards with enhanced styling */
```

```
/* Enhanced voice button */
```

```
/* Sidebar enhancements */
```

```
/* Section titles */
```

```
/* Detail cards */
```

```
if 'audio_path' not in st.session_state:

# Main header with plant theme (translated)
s

model = load_model()
c
if model is None or not class_names:
```

```
uploaded_file = st.file_uploader(trans['upload'], type=["jpg", "jpeg", "png"])
if uploaded_file is not None:
    with col1:




    with col2:


        # ? CONFIDENCE METER CARD
                                                                                                unsaf




        # Color-coded confidence label
```

```
                st.markdown(f'<div style="text-align: center; font-size: 1.2em; margin: 15px
0;">{confidence_label
                                                                                                }</di

            # ?? SEVERITY LEVEL CARD

                                                                                                unsaf

                # Create comprehensive voice text with translations

                                                                                                disea

                col_speak, col_pause = st.columns([1, 1])
```

```
            if st.session_state.audio_path and not st.session_state.audio_paused:

        # ? DETAILED ANALYSIS SECTION
                                                                                                    unsaf

            if disease_info_translated:
                                                                                                    unsaf

                                                    st.markdown(f'<div  class="info-card"><strong>?
{trans["description"]}:</strong><br>{diseas
                                                                                                    e_inf
                                                                                                    unsaf

                st.markdown('</div>', unsafe_allow_html=True)

                # ? TREATMENT CARD
                                                                                                    unsaf
```

========== ? AI CHAT ASSISTANT SECTION ==========

? **Description:**

? **Causes:**

? **Treatment Options:**

?? **Prevention Tips:**

? **Description:**

? **Causes:**

? **Immediate Treatment:**

?? **Prevention Strategies:**

- Use

? **Your plant is disease-free! Here's how to keep it healthy:**

? **Care Requirements:**

- Con

?? **Prevention Measures:**

- Mon

? **Monitoring Tips:**

- Che

? **Boost Plant Immunity:**

- Use

? **Treatment Categories:**

Chemical Fungicides:

- Cop

Organic Solutions:

- Nee

Application Tips:

- Spr

?? **Important:**

- Alt

?? **Top Prevention Strategies:**

Cultural Practices:

- Cro

Resistant Varieties:

- Cho

Environmental Control:

- Mai

Regular Monitoring:

- Wee

Soil Health:

- Tes

? **Basic Facts:**

- Clo

? **Key Features:**
- Sur

? **Fun Facts:**
- A d

? **Basic Facts:**
- Sec

? **Key Features:**
- Hot

? **Fun Facts:**
- Ven

? **Basic Facts:**
- Thi

? **Key Features:**
- 71%

? **Fun Facts:**
- Onl

? **Basic Facts:**

- Fou

? **Key Features:**

- Two

? **Fun Facts:**

- Mos

? **Basic Facts:**

- Fif

? **Key Features:**

- 79

? **Fun Facts:**

- Rad

? **Basic Facts:**

- Six

? **Key Features:**

- 146

? **Fun Facts:**

- Rin

?? **Basic Facts:**

- Sev

?? **Key Features:**

- Ice

? **Fun Facts:**

- Ext

? **Basic Facts:**

- Eig

? **Key Features:**

- Str

? **Fun Facts:**

- Fur

I can help you with:

? **Plant Knowledge:**
- Ear

? **Space Knowledge:**
- Mer

Try asking about:
- "Wh

What would you like to know? ?

```
def run():
                                                
```